ct: DSP
ner: Dr. Samir Kamal

Term: First Acd. Year: 2013/2014
Year: 5th Comm. Time Allowed: 75 min.

M. T. Exam (Model - A)

stion 1

- Compare between deterministic and random signals
- Derive the relation between the Fourier transform and the Discrete Fourier Transform algorithms.
- In detail, what are the main characteristics of the continuous time sinusoidal signals?
- Determine which of the following signals are periodic, and then compute their fundamental period.

(i) $X(n)= \cos(3n\pi)$ (ii) $x_a(t) = 3\cos(5t + \pi/6)$

stion 2

n the two set of samples $x_1(n) = \{1, 0, 1, 0\}$ and $x_2(n) = \{0, 2, 0, 1\}$.

- Calculate the frequency components of $x_1(n)$, X1(K) using the DFT algorithm.
- Calculate the frequency components of $x_2(n)$, X2(K) using the FFT algorithm.
- Using X1(K) and $x_1(n)$, verify the Parseval's theorem.
- Find the computational saving of using FFT algorithm instead of using the DFT, algorithm for calculating X1(k).

Subject: DSP
Examiner: Dr. Samir Kamal

Term: First Acd. Year: 2013/2014
Year: 5th Comm. Time Allowed: 78 min.

M. T. Exam (Model - B)

Question 1

a- Show that the DFT, X (k), of a sequence x (n) is periodic.
b- In detail, what are the main characteristics of the continuous time sinusoidal signals?
c- Deduce the relation between the DFT algorithm and the Fourier transform.
d- Determine which of the following signals are periodic, and then compute their fundamental period.
(i) $X(n) = \cos(0.01\pi n)$ (ii) $x_a(t) = 3\cos(5t + \pi/3)$

Question 2

For the signal $x(t) = \cos(200\pi t)$ to be sampled with sampling frequency F_s = 400Hz.

a- Find the discrete time sequence x(n) of $x(t)$ then calculate its fundamental period, N_p.
b- Find the first four samples values of $x(n)$ then calculate its freq. components, X(k) using the DFT algorithm.
c- Using the results obtained in part (b) verifies the Parseval's theorem.
d- Find the computational saving of using FFT algorithm instead of using the DFT, algorithm for calculating X(k).

Subject: DSP
Examiner: Dr. Samir Kamal

Term: First
Year: 5th Comm.

Acd.Year: 2011/2012
Time Allowed: 1 Hour

DSP - Mid-Term Exam (Model-C)

Q # 1 Prove that the DFT, X (k), of a sequence x (n) is periodic.

Q # 2 In detail, Explain the sampling process used in A/D converter.

Q # 3 For the following sequence, x (n): x (n) = {2, 1, 0, 1}:

a- Using the direct method find the frequency components of the sequence X (k) and check your results by calculating its IDFT.

c- Using your results, verify Parseval's theorem.

d- Plot the amplitude and the phase spectrums.

Subject: DSP
Examiner: Dr. Samir Kamal

Term: First Acd. Year: 2012/2013
Year: 5th Comm. Time Allowed: 1 Hour

Mid-Term Exam (Model-B)

Question 1

a- Show that the DFT, X (k), of a sequence x (n) is periodic.

b- In detail, what are the main characteristics of the continuous time sinusoidal signals?

Question 2

For the signal $x(t) = \cos(200\pi t)$ to be sampled with sampling frequency F_S = 400Hz.

a- Find an expression for its samples x(n) then calculate its fundamental period, N_P.

b- Find the samples values of the first cycle in *x(n)* then plot them at the same figure with $x(t)$.

c- Calculate the DFT, X(K), for *x(n)* obtained in part (b).

d- Using the results obtained in part (b) and part (c) verifies the Parseval's theorem.

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Term: First
Year: 5th Comm.

Acd.Year: 2011/2012
Time Allowed: 1 Hour

DSP - Mid-Term Exam (Model-B)

Q # 1 Deduce the basic relation between the Fourier transform and Discrete Fourier transform.

Q # 2 In detail, Explain the quantization process used in A/D converter.

Q # 3 For the following sequence, x (n): x (n) = {0, 1, 0, 1}:

a- Using the direct method find the frequency components of the sequence X (k) and check your results by calculating its IDFT.

c- Using your results, verify Parseval's theorem.

d- Plot the amplitude and the phase spectrums.

Subject: DSP
Examiner: Dr. Samir Kamal

Term: First Acd. Year: 2013/2014
Year: 5th Comm. Time Allowed: 75 min.

M. T. Exam (Model - A)

Question 1

a- Compare between deterministic and random signals

b- Derive the relation between the Fourier transform and the Discrete Fourier Transform algorithms.

c- In detail, what are the main characteristics of the continuous time sinusoidal signals?

d- Determine which of the following signals are periodic, and then compute their fundamental period.

(i) $X(n)= \cos(3n\pi)$ (ii) $x_a(t) = 3\cos(5t + \pi/6)$

Question 2

Given the two set of samples $x_1(n) = \{1, 0, 1, 0\}$ and $x_2(n) = \{0, 2, 0, 1\}$.

a- Calculate the frequency components of $x_1(n)$, X1(K) using the DFT algorithm.

b- Calculate the frequency components of $x_2(n)$, X2(K) using the FFT algorithm.

c- Using X1(K) and $x_1(n)$, verify the Parseval's theorem.

d- Find the computational saving of using FFT algorithm instead of using the DFT, algorithm for calculating X1(k).

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Mid-Term Exam (Model-A)

Question 1

a- Explain using block diagram the stages of analog- to- digital conversion process.
b- Derive the relation between the Fourier transform and the Discrete Fourier Transform.

Question 2

Given the analog signal: $x(t) = 5\cos(2000\pi t) + \sin(4000\pi t)$.

a- Using the sampling frequency F_S = 3 kHz, could the original signal be reconstructed successfully from its samples?
b- Using F_S = 8 kHz, find the discrete time sequence x(n) of $x(t)$ and calculate the values for the first four samples.
c- Find the FFT, X(K), for x(n) obtained in part (b).
d- Using the results obtained in part (b) and part (c) verifies the Parseval's theorem.

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Subject: DSP
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M. T. Exam (Model - B)

Question 1

a- Show that the DFT, X (k), of a sequence x (n) is periodic.
b- In detail, what are the main characteristics of the continuous time sinusoidal signals?
c- Deduce the relation between the DFT algorithm and the Fourier transform.
d- Determine which of the following signals are periodic, and then compute their fundamental period.
(i) $X(n)= \cos(0.01n\pi)$ (ii) $x_a(t) = 3\cos(5t + \pi/3)$

Question 2

For the signal $x(t) = \cos(200\pi t)$ to be sampled with sampling frequency F_s = 400Hz.

a- Find the discrete time sequence x(n) of $x(t)$ then calculate its fundamental period, N_p.
b- Find the first four samples values of $x(n)$ then calculate its freq. components, X(k) using the DFT algorithm.
c- Using the results obtained in part (b) verifies the Parseval's theorem.
d- Find the computational saving of using FFT algorithm instead of using the DFT, algorithm for calculating X(k).

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Term: First Acd. Year: 2012/2013
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Mid-Term Exam (Model-C)

tion 1

Define each of the following terms using examples, figures and equations when necessary:
Signal – System – Signal Processing – Sampling – Quantization.
In detail, what are the main characteristics of the discrete time sinusoidal signals?

tion 2

the two set of samples $x_1(n) = \{-1 , 0 , 2 , 0\}$ and $x_2(n) = \{0 , -2 , 1 , 0\}$.
Calculate the DFT, X1(K), for $x_1(n)$.
Calculate the FFT, X2(K), for $x_2(n)$.
Using X1(K) and $x_1(n)$ verify the Parseval's theorem.

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Examiner: Dr. Samir Kamal

Term: First Acd. Year: 2013/2014
Year: 5th Comm. Time Allowed: 75 min.

M. T. Exam (Model - A)

Question 1

a- Compare between deterministic and random signals
b- Derive the relation between the Fourier transform and the Discrete Fourier Transform algorithms.
c- In detail, what are the main characteristics of the continuous time sinusoidal signals?
d- Determine which of the following signals are periodic, and then compute their fundamental period.
(i) $X(n)= \cos(3n\pi)$ (ii) $x_a(t) = 3\cos(5t + \pi/6)$

Question 2

Given the two set of samples $x_1(n) = \{1, 0, 1, 0\}$ and $x_2(n) = \{0, 2, 0, 1\}$.
a- Calculate the frequency components of $x_1(n)$, X1(K) using the DFT algorithm.
b- Calculate the frequency components of $x_2(n)$, X2(K) using the FFT algorithm.
c- Using X1(K) and $x_1(n)$, verify the Parseval's theorem.
d- Find the computational saving of using FFT algorithm instead of using the DFT, algorithm for calculating X1(k).

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Term: First Acd. Year: 2013/2014
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Subject: DSP
Examiner: Dr. Samir Kamal

M. T. Exam (Model - B)

Question 1

a- Show that the DFT, X (k), of a sequence x (n) is periodic.
b- In detail, what are the main characteristics of the continuous time sinusoidal signals?
c- Deduce the relation between the DFT algorithm and the Fourier transform.
d- Determine which of the following signals are periodic, and then compute their fundamental period.
(i) $X(n)= \cos(0.01n\pi)$ (ii) $x_a(t) = 3\cos(5t + \pi/3)$

Question 2

For the signal $x(t) = \cos(200\pi t)$ to be sampled with sampling frequency F_S = 400Hz.

a- Find the discrete time sequence x(n) of $x(t)$ then calculate its fundamental period, N_P.
b- Find the first four samples values of $x(n)$ then calculate its freq. components, X(k) using the DFT algorithm.
c- Using the results obtained in part (b) verifies the Parseval's theorem.
d- Find the computational saving of using FFT algorithm instead of using the DFT, algorithm for calculating X(k).

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Term: First Acd. Year: 2012/2013
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Mid-Term Exam (Model-C)

estion 1

- Define each of the following terms using examples, figures and equations when necessary:
  Signal – System – Signal Processing – Sampling – Quantization.
- In detail, what are the main characteristics of the discrete time sinusoidal signals?

estion 2

en the two set of samples $x_1(n) = \{-1 , 0 , 2 , 0\}$ and $x_2(n) = \{0 , -2 , 1 , 0\}$.

- Calculate the DFT, X1(K), for $x_1(n)$.
- Calculate the FFT, X2(K), for $x_2(n)$.
- Using X1(K) and $x_1(n)$ verify the Parseval's theorem.

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Term: First
Year: 5th Comm.

Acd.Year: 2011/2012
Time Allowed: 1 Hour

DSP - Mid-Term Exam (Model-B)

Q # 1 Deduce the basic relation between the Fourier transform and Discrete Fourier transform.

Q # 2 In detail, Explain the quantization process used in A/D converter.

Q # 3 For the following sequence, x (n): x (n) = {0, 1, 0, 1}:

a- Using the direct method find the frequency components of the sequence X (k) and check your results by calculating its IDFT.

c- Using your results, verify Parseval's theorem.

d- Plot the amplitude and the phase spectrums.

Quiz (1)

Q# 1

1- Define the following items: signal , system , signal processing and sampling condition.

2- Draw the block diagrams of both A/D converter& DS processor.

3- In detail , Explain the quantization process used in A/D converter.

4- Specify the fundamental period(N) of the signal $X(n)=2\sin(3\pi n/105)$.

Q#2 in quantization of analog sinusoidal signal , Drive an expression for the signal to quantization noise power ratio in dB, SQNR(dB) .

Q#3 Given the following analog signal $Xa= 10\cos(1500\pi t)$

1- if it sampled with Fs=3000 samples/ sec , is it possible to reconstruct that signal back from its samples successfully?

2- Calculate the sanples values during one cycle of the signal and sketch both analog and discrete signals on one graph

3-Using the direct method find the DFT of those samples obtained in part (b).

4-Check your results by calculating the inverse DFT(IDFT).

5-Compute the computational saving when we use the direct method or FFT.

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Mid-Term Exam (Model-B)

Question 1

a- Show that the DFT, X (k), of a sequence x (n) is periodic.

b- In detail, what are the main characteristics of the continuous time sinusoidal signals?

Question 2

For the signal $x(t) = \cos(200\pi t)$ to be sampled with sampling frequency F_S = 400Hz.

a- Find an expression for its samples x(n) then calculate its fundamental period, N_P.

b- Find the samples values of the first cycle in *x(n)* then plot them at the same figure with $x(t)$.

c- Calculate the DFT, X(K), for *x(n)* obtained in part (b).

d- Using the results obtained in part (b) and part (c) verifies the Parseval's theorem.

Model- 1- Corrective - DSP-M T Exam- Fall 2021 - 26/ 12/ 2021 (40 mints)

Question 1 [15 Points]

For the signal $x(t) = 2\cos(400\pi t)$ to be sampled with sampling frequency F_S = 400Hz.

(i) Find an expression for its corresponding sequence, x(n). Then calculate its fundamental period N_P.

(ii) Find the samples values of the first cycle in x(n) then plot them at the same figure with $x(t)$.

Question 2 [15 Points]

For the following sequence, x (n): x (n) = {1, ↓, 0, 1}:

(i) Using the DFT method find the frequency components, X(k) and check your results by calculating its IFFT method.

(ii) Using your results, verify Parseval's theorem.

(iii) Plot the amplitude and the phase spectrums.